PeerEval Aligning LLM Peer Judgments with Human Preferences

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Research objective

We propose to develop and validate human-calibrated peer evaluation: a method that combines judgments from multiple LLMs with a small calibration set of human preferences to produce reliable model rankings. Arena has documented self-preference, family preference, and substantial disagreement between AI judges and human voters [3]. Its AutoEval system demonstrates the value of fast automated ratings grounded in human feedback [4]. These findings motivate a specific question: when can a panel of LLMs provide trustworthy preference estimates, and when does apparent consensus reflect a shared bias? Our goal is to improve the reliability and cost of automated evaluation while keeping human preferences as the external reference.

Foundation and proposed advance

Our PRD work introduced Peer Rank, which iteratively weights each model as a judge according to how highly its own answers are rated by peers [1]. We have implemented this approach in PeerEval, with an open-source evaluation package and a public leaderboard; the initial release contains seven models, three benchmarks, and 40,156 pairwise judgments [2]. This infrastructure makes the proposed study immediately executable. The unresolved limitation is that answering well does not necessarily imply judging in accordance with human preferences. The proposed advance is to use peer competence as a prior, then calibrate judge influence against limited human feedback and explicitly test shared errors.

Research approach

Calibrate peer judgments to human preferences. Each model will judge anonymous response pairs in both display orders. On a separate calibration split, we will fit a regularized, nonnegative mixture of judge votes, using PRD weights as a prior. The mixture will retain distinct outcomes for an A win, a B win, a tie, and both responses being bad. We will compare self-judgment exclusion and family-aware weighting, and penalize redundant influence from judges whose errors on the calibration set are highly correlated. Label-budget experiments will test whether peer weights provide a useful starting point compared with uniform weights or weights learned only from human labels. No held-out audit labels will enter weight fitting or model prompts.

Spend additional evaluation effort where rankings are uncertain. A sequential evaluation policy will first query a small panel, then request an additional judge from another family or a human vote when disagreement, model novelty, or uncertainty could change a consequential ranking comparison. It will retain random exploration and use inverse-probability weighting to account for selective comparisons. We will compare this policy with random allocation and a complete judge panel at matched dollar and human-label budgets. Independent peer judgments remain the default; agreement alone will never be treated as evidence of human alignment.

Evaluation and success criteria

We will target approximately 10,000 permitted Arena text battles spanning coding assistance, knowledge-intensive requests, and creative writing, with six to eight judges from multiple model families. Public datasets support the core study; subject to agreement, recent Arena battles would enable stronger tests on newly introduced models and changing preferences. We will create separate calibration, acquisition, and untouched audit pools grouped by prompt, then test later time windows and held-out answer-model families. Human-label budgets of 0, 50, 100, 250, and 500 comparisons will count both calibration and acquired votes. A small set of repeated human annotations will quantify preference ambiguity.

Baselines will include frontier single judges, uniform panels, original PRD, human-calibrated mixtures without the PRD prior, and family-diverse panels. If feasible, we will work with Arena to include its reward model on the same held-out comparisons. Primary measures are preference log loss and Kendall correlation with human model rankings; secondary measures cover tie handling, self- and family-preference relative to human votes, and residual length or formatting preferences. Prompt-level bootstrap intervals will refit calibration and rankings. Success means better held-out calibration than uniform panels and uncalibrated PRD at equal cost, with a target of at least 25% fewer judge calls than a complete panel at matched ranking agreement. We will report failures and uncertainty alongside gains.

Expected outputs and project schedule

Over 12 months, we will deliver a research paper, an open-source extension of PeerEval for calibrated weighting and sequential evaluation, and a reproducible benchmark report with evaluation costs, bias audits, and ranking uncertainty. Public releases will include permitted judgments, configurations, and analysis code; restricted Arena data will remain subject to its access agreement. Months 1 to 3 will establish data splits and baselines; months 4 to 6 will develop and assess calibrated weighting; months 7 to 9 will test sequential allocation and transfer; months 10 to 12 will complete independent audits, the paper, and the code release. We welcome collaboration with Arena researchers and a joint public explanation of the findings.

Requested support and research value

We request a $50,000 research gift: approximately $36,000 for partial graduate research support, including applicable fringe and tuition; $10,000 for model API access and computing; and $4,000 for repeated human annotations and targeted audits. The proposed allocation will follow institutional requirements. Arena would gain evidence about when peer evaluation can supplement human voting and AutoEval, how judge-pool composition affects validity, and which comparisons most need additional human feedback. The study remains useful if a human-calibrated reward model outperforms peer aggregation: it will identify the conditions under which peer signals add information and where they should receive little weight.

References

[1] Li R, Patel T, Du X. PRD Peer Rank and Discussion Improve Large Language Model based Evaluations. TMLR, 2024. Paper.

[2] PeerEval. Leaderboard, project blog, and open-source evaluation package. 2026. Project.

[3] Arena Team. Self-Preference of AI Judges. September 30, 2026. Article.

[4] Arena Team. Introducing AutoEval to the Arena leaderboards. July 30, 2026. Article.